

Tucker Johnson

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Summary of Qualifications

- Mechatronics Engineering student with interests in controls, embedded systems, robotics, and autonomous technologies
- Experience designing mechanical systems and electronics using SolidWorks and KiCad for engineering work
- Hands-on experience building, testing, troubleshooting, and validating electromechanical systems through iterative design
- Experience developing full-stack software and embedded applications using Python, C++ and AI development tools
- Strong communication and teamwork developed through technical collaboration and documentation of engineering projects

Education

University of Waterloo, Bachelor of Applied Science - Mechatronic Engineering

Sept 2025 – June 2030

- **Awards:** President's Scholarship of Distinction 2025, 4.0 GPA

Technical Skills

Mechanical Design and Analysis: SolidWorks, AutoCAD, Onshape, KiCad, SimScale, HydroD, GD&T, Microsoft Excel

Programming & Software: C++, Python, JavaScript, Node.js, FastAPI, React, Git, SQLite, Linux, MATLAB

Experience

Mechanical Engineering Intern with JASCO Applied Sciences & COVE

May 2026 - August 2026

- Designed, manufactured, and validated test fixtures for battery packs, SD cards, and hydrophone calibration using SolidWorks, reducing mobilization time by 25%
- Developed test procedures and conducted mechanical testing and failure analysis of bolts, supporting design improvements for underwater housings.
- Developed test procedures and characterized pressure relief valves (PRVs) through experimental testing and analysis, improving the safety and reliability of underwater housings.
- Redesigned and standardized the cable management system for over 80 cables, improving inventory tracking, procurement efficiency, and cable identification by length and application.
- Maintained and organized SolidWorks PDM Vault by updating mechanical parts, assemblies, and engineering drawings, ensuring accurate documentation and long-term design traceability.
- Assembled buoys, acoustic housings, and moorings while supporting packaging, and deployment preparation for the REPMUS 2026 international naval exercise.

Waterloo Arial Robotics Group

- Designed and developed embedded software for a drone altimeter using C++, implementing sensor integration and flight-state logic on a microcontroller.
- Designed electronic schematics and PCBs in KiCad, selecting components and optimizing circuit layouts for reliability.
- Manufactured and assembled custom PCBs through soldering, inspection, and hardware integration to produce functional flight-ready electronics.
- Tested, debugged, and validated embedded hardware and software using oscilloscopes, multimeters, and serial debugging to ensure reliable system performance.

Volleyball Volunteer, Volleyball Nova Scotia

Present

- Managed operations of a two-court facility supporting 500+ participants, coordinating schedules, teams, and logistics
- Demonstrated leadership, organization, and clear communication in a high-responsibility environment

Certifications and Awards

Certifications

- Waterloo University SolidWorks Certification
- University of Waterloo Shop Safety Certification
- Drone Pilot Certificate - Basic Operations & RPAS Advanced Pilot Course
- French language: 18 on OPI test

Awards

- 2025 Canada Cup Maple Award Winner: Best teammate on NS Male Canada Games Volleyball Team